

**CBSE Test Paper 04**  
**Chapter 10 Mechanical Properties of Fluids**

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1. A boy is carrying a bucket of water in one hand with a 0.05 kg cork piece in it. After sometime he takes out the cork piece and holds it in the other hand. The boy will carry **1**
  - a. Same load as before
  - b. Less load than before
  - c. More load than before
  - d. Either less or more load depending on the density of plastic
2. Soldering of two metals possible because of the property of **1**
  - a. osmosis
  - b. cohesion
  - c. surface tension
  - d. viscosity
3. A liquid drop of radius  $R$  is broken up into  $N$  small droplets. The work done is proportional to (take  $N^{\frac{1}{3}} \gg 1$ ) **1**
  - a.  $N^{1/3}$
  - b.  $N$
  - c.  $N^{-2/3}$
  - d.  $N^{-0.5}$
4. If there would be smaller gravitational effects, which of the following forces do you think would alter in some respect? **1**
  - a. Nuclear forces
  - b. Electrostatic forces
  - c. Viscous forces
  - d. Buoyant forces
5. What fraction of the volume of solid piece of metal of sp gr 7.2 floats above the surface of a container of mercury of sp. gr. 13.6? **1**
  - a. 0.78
  - b. 0.47
  - c. 0.53

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d. 0.75

6. Define viscosity? 1
  7. On what factors does the critical speed of fluid flow depend? 1
  8. Water rises in a capillary tube but mercury falls in the same tube. Why? 1
  9. Two vessels have the same base area but different shapes. The first vessel takes twice the volume of water that the second vessel requires to fill upto a particular common height. Is the force exerted by the water on the base of the vessel the same? Why do the vessels filled to same height give different reading on weighing scale. 2
  10. The sap in trees, which consists mainly of water in summer, rises in a system of capillaries of radius  $r = 2.5 \times 10^{-5}$  m. The surface tension of sap is  $T = 7.28 \times 10^{-2} \text{ Nm}^{-1}$  and the angle of contact is  $0^\circ$ . Does surface tension alone account for the supply of water to the top of all trees? 2
  11. A metal plate  $5 \text{ cm} \times 5 \text{ cm}$  rests on a layer of castor oil 1 mm thick whose co-efficient of viscosity is  $1.55 \text{ Nsm}^{-2}$ . What is the horizontal force required to move the plate with a speed of  $2 \text{ cm s}^{-1}$ ? 2
  12. A vessel filled with water is kept on a weighing pan and the scale adjusted to zero. A block of mass  $M$  and  $\rho$  density is suspended by a mass less spring of spring constant  $k$ . This block is submerged inside into the water in the vessel. What is the reading of the scale? 3
  13. What is terminal velocity and derive an expression for it? 3
  14. The flow rate of water from a tap of diameter 1.25 cm is 0.48 L/min. The coefficient of viscosity of water is  $10^{-3} \text{ Pa-s}$ . After some time the flow rate is increased to 3 L/min. Characterise the flow for both the flow rates. 3
  15. In a test experiment on a model aeroplane in a wind tunnel, the flow speeds on the upper and lower surfaces of the wing are  $70 \text{ ms}^{-1}$  and  $63 \text{ ms}^{-1}$  respectively. What is the lift on the wing if its area is  $2.5 \text{ m}^2$ ? Take the density of air to be  $1.3 \text{ kg m}^{-3}$ . 5
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